

Thomas Robert Butler IV - Senior AI/ML Engineer

trbiv@proton.me | Github | LinkedIn

Summary

- **Education:** University of Iowa, BSE Computer Science and Engineering
- **Languages:** Javascript/Typescript, Go, Rust, Java, Kotlin, Solidity, Cairo, Python

Work Experience

Senior Software Engineer – GFX Labs (October 2025 - Present)

- Led perps trading platform from ideation to release, delivering Hyperliquid integration with 100+ markets and isolated/cross margin.
- Built full-stack infrastructure across React/TypeScript frontend and Go backend, orchestrating signing flows and order submission with Temporal.
- Led World App miniapp from ideation to release, building end-to-end for token swaps on World Chain with Permit2 proxy, custom swap routing, and MiniKit v2.
- Headed creation and integration of design system component library across both main trading application and miniapp, using Storybook, Chromatic, and in-house visual regression testing.
- **Tools Used:** React, TypeScript, Go, Temporal, Viem, Wagmi, gRPC/Protobuf, Grafana, PostHog, Kubernetes, Storybook, Playwright, Chromatic

Senior Software Engineer – Nakamoto LLC (January 2025 - June 2025)

- Led team of two engineers building AI inference infrastructure and data pipelines for Bittensor subnets.
- Fine-tuned tool-calling LLMs, outperforming SOTA models on Berkeley Function Calling Leaderboard with top 10% subnet performance.
- Unified physical datacenter and cloud servers into a single Kubernetes cluster delivering auto-scaled AI inference.
- Built concurrent microservice architecture in Python and Go for large-scale dataset harvesting across Reddit, Twitter, and YouTube.
- **Tools Used:** Python, Golang, Rust, Linux, Kubernetes, Postgres, Redis, Docker, Git/Github, Huggingface, AWS, PyTorch, NumPy, Sglang, Vllm, AI Agents, ElizaOS, Daydreams

Mobile Software Engineer – Tholos (October 2023 - January 2025)

- Owned iOS and Android mobile applications for Tholos MPC Crypto Wallet in React Native, securing over \$35 Million in digital assets.
- Built full-stack features spanning WebSocket connections, AWS secure enclave, and multi-chain key management across Ethereum, Bitcoin, Solana, and Cosmos.
- Implemented fault-tolerant key backup and recovery achieving 100% recovery rate across distributed mobile devices.
- **Tools Used:** React Native, Native Wind, Expo, JavaScript/Typescript, Bash, Python, Go, Rust, Postgres, Redis, AWS, Docker, Git/Github, FastAPI

Project Highlights

AI Smart Contract Vulnerability Detection - Github

- Supervised and Unsupervised methods to detect potential vulnerabilities in smart contracts.
- Methods included analyzing bytecode directly by constructing a 'sliding window' of relevant opcodes, as well converting bytecode into RGB images, both of which were analyzed using neural networks.
- **Tools Used:** Python, PyTorch, Jupyter Notebook, Solidity, Slither